Zewail City School of Science and Technology		Group: 4th Year
School of Computational Sciences and		Course Name: Reinforcement Learning
Artificial Intelligence		Course Code: DSAI 402
Date: January 19, 2026		Time Allowed: 1.5 hours
Total Marks: 40		No. of Pages: 6

Answer the questions in the spaces provided on the question sheets.

GROUP A

Name: _____

Serial No.: _____

Question	Points	Score
1	6	
2	3	
3	10	
4	10	
5	7	
6	2	
7	2	
Total:	40	

In **ALL** questions, you need to show **all the steps** required to obtain your answers.

Do not answer in this page!

Q1 Given the following three episodes recorded from an agent in a gridworld. Transitions are written as (s_t, a_t, r_{t+1}).

- **Episode 1:** s_1, right, -3, s_2, down, 3, s_4, terminated, $+4$
- **Episode 2:** s_1, down, 2, s_2, right, 2, s_4, terminated, $+4$
- **Episode 3:** s_1, right, -3, s_2, up, -2, s_1, right, -1, s_2, down, 3, s_4, terminated, $+4$

(a) (6 points) Compute the policy using the Monte-Carlo first visit algorithm. Then answer the following

i. what is the greedy policy $\pi(s_1)$?

a) right b) down c) up

ii. what is the greedy policy $\pi(s_2)$?

a) right b) down c) up

Q2 (3 points) Scenario: Imagine you are training a Deep Q-Network (DQN) to play a video game. You decide to simplify the algorithm to save memory and computation. You make two specific changes:

- Change A: You remove the Target Network. So, you use the current weights θ to calculate both the current Q-value and the "next state" Q-value in the loss function.
- Change B: You remove the Experience Replay Buffer. Instead of sampling random batches from the past, you update the network using only the very last transition (s, a, r, s') the agent just experienced.

Explain the issue with each change.

Q3 Below is the state-value estimates (V) for a 4×4 grid. The agent can move Up, Down, Left, or Right. If the agent moves into a wall, it stays in the same square. For simplicity, assume $\gamma = 1$ and the reward $r = 0$ for all transitions unless stated otherwise.

$V_A =$

	C_1	C_2	C_3	C_4
R_1	-2	2	5	9
R_2	-3	-2	2	5
R_3	-3	-3	-2	2
R_4	-4	-3	-3	-2

$V_B =$

	C_1	C_2	C_3	C_4
R_1	3	-4	-4	-6
R_2	5	3	-4	-4
R_3	6	5	3	-4
R_4	10	6	5	3

(a) (2 points) Draw the policy for advisor V_A?

	C_1	C_2	C_3	C_4
R_1				
R_2				
R_3				
R_4				

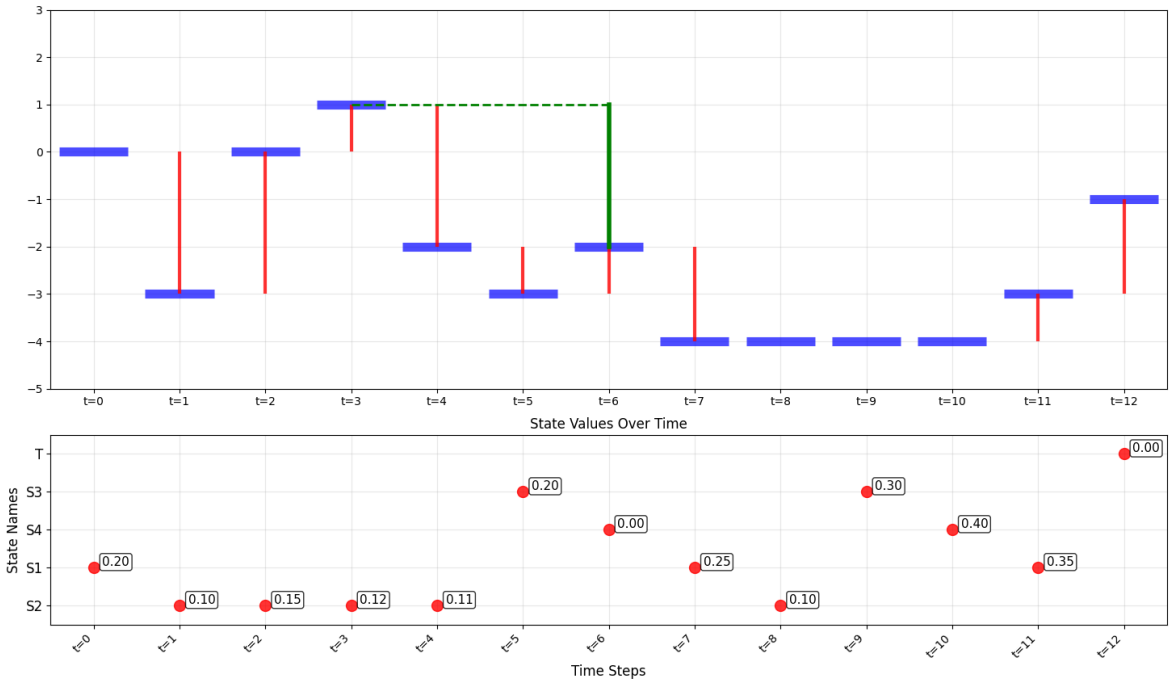
(b) (2 points) What would be the best action for the state (R_3, C_3) for V_B. Justify your answer.

(c) (4 points) Neither of the two advisors (V_A and V_B) is perfect. **Without running any iterative algorithms**, create a single policy π_{new} that is better than the greedy policy of V_A and the greedy policy of V_B. Justify your answer.

	C_1	C_2	C_3	C_4
R_1				
R_2				
R_3				
R_4				

(d) (2 points) If there is a hidden reward of $+10$ at (R_1, C_4) AND a reward of $+10$ at (R_4, C_1), which of the three policies (V_A, V_B, or your New Policy) is most likely to find either reward regardless of where the agent starts? Why?

Q4 Based on the following figure



(a) (2 points) What does the horizontal blue bar represent?

(b) (2 points) What does the vertical red bar represent?

(c) (2 points) What does the green vertical line (at time $t = 6$) represent?

(d) (2 points) Is the figure valid for any value of γ? Why?

(e) (2 points) What would be the value of the state S_2 at time $t = 3$ after one iteration of the 3-step TD algorithm. Assume $\alpha = 0.3$.

Q5 Below is the current Q-table for an agent. Assume that the agent is at the state S_1 and takes action Right and receives a reward of -1. It arrives at the state S_2. In S_2, the agent's ϵ-greedy policy chooses the action Up (perhaps due to exploration, even though Up does not have the highest value in the table). Assume that $\gamma = 0.9$ and $\alpha = 0.5$.

$Q(s, a) =$

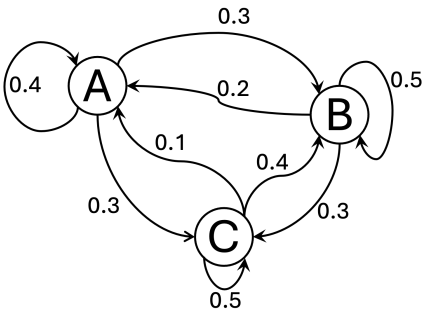
	Up	Down	Right
S_1	0	0	2
S_2	-10	0	5

(a) (2 points) Calculate the new value for $Q(S_1, Right)$ using SARSA.

(b) (2 points) Calculate the new value for $Q(S_1, Right)$ using Q-Learning.

(c) (3 points) Compare the two values. How does this reflect the difference between "on-policy" and "off-policy" learning?

Q6 Consider a Markov chain with three states as shown in the following figure.



(a) (2 points) Calculate the probability of transitioning from state A to state C in exactly two steps.

Q7 (2 points) Consider the following state-value diagram. If the Agent's policy π is to choose action a_1 with a probability of 0.2 and action a_2 with a probability of 0.8 , and we assume the discount factor $\gamma = 0$ and the next states have a value of 0, what is the value of the starting state $V(s)$? Show all your steps.

